ACTIVITY-BASED LEARNING LAB REPORT

Department of Computer Science & Engineering | Web Development Project Suite

Student Name: D. DHATTASRI SIVA SAI

Roll Number: 26000017

Branch & Section: CSE - Section 1

Academic Year: 2025–2026

1. EXECUTIVE OVERVIEW

Lab Objectives & Technical Scope

This report documents the design, implementation, and visual verification of three web applications developed as part of the Web Development Activity-Based Learning coursework. The project demonstrates progressive mastery of standard HTML5 semantic structuring, CSS3 layout engines (Flexbox, styled grids), form validations, and user-centric UI design patterns.

2. CASE STUDY 1: PERSONAL PORTFOLIO WEBPAGE (CASE1)

Case 1 - Personal Portfolio Website

Basic HTML5 & CSS Layout

Description: A personal brand website presenting student credentials, technical skill sets, academic history, and foundational projects. Built with clear semantic blocks (`<header>`, `<nav>`, `<section>`, `<footer>`) and structured CSS styling.

- **Navigation Bar:** Smooth anchor link navigation to page sections.
- **Skills Matrix:** Styled inline tags using custom background badges (`#dbeafe`).
- **Projects Showcase:** Distinct card modules highlighting academic works.

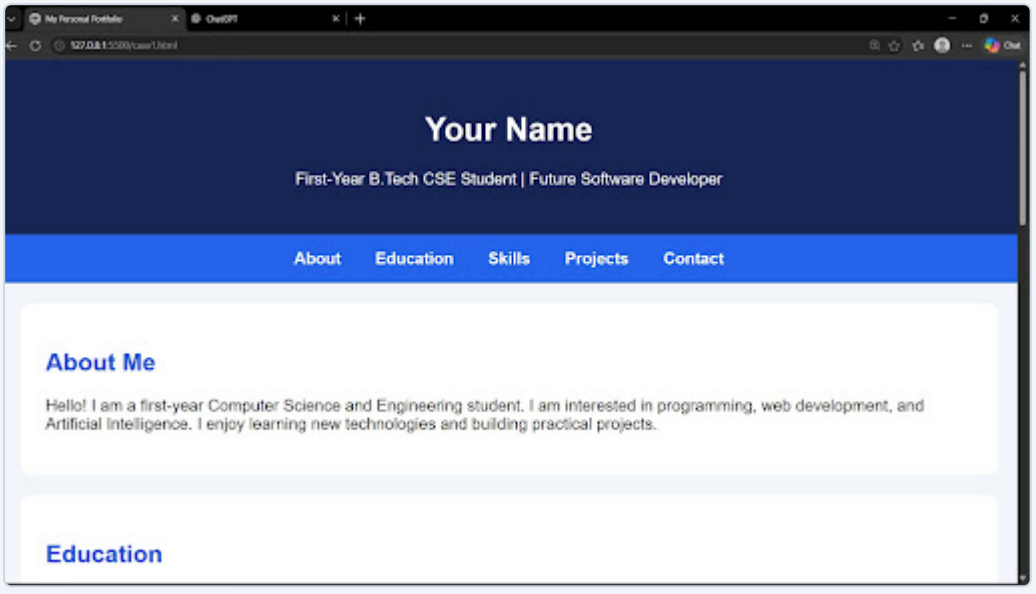


Figure 1.1: Web Preview Output for Case 1 (Personal Portfolio Webpage)

3. CASE STUDY 2: SWEET TREATS BAKERY WEBPAGE (CASE9)

Case 9 - Sweet Treats Bakery Page

E-Commerce & CSS Grid Layout

Description: A commercial bakery storefront featuring a multi-item product showcase. Utilizes CSS Flexbox/Grid for responsive product cards, hover states, pricing callouts, promotional banners, and location contact details.

- **Visual Product Cards:** Image containers with explicit dimensions and uniform spacing.
- **E-Commerce Styling:** Highlighted pricing tags in bold green (#218044) and styled order buttons.
- **Promotions Panel:** Warm, themed callout section for seasonal discounts.

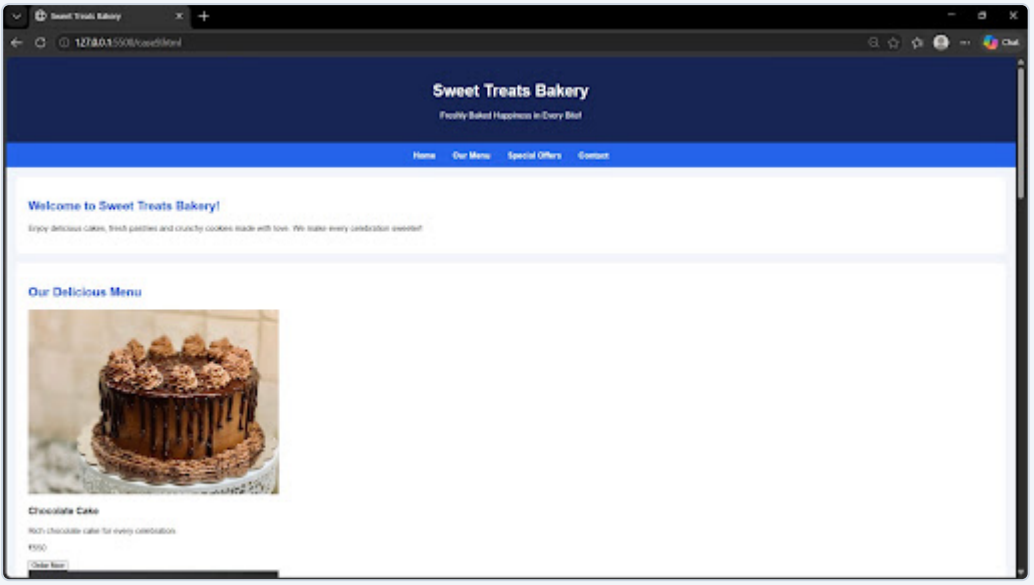


Figure 2.1: Web Preview Output for Case 9 (Sweet Treats Bakery Website)

4. TECHNICAL IMPLEMENTATION SNAPSHOT (CASE9.CSS)

```
      .products { display: flex; flex-wrap: wrap; justify-content: center; gap: 20px; } .item { bac
kground-color: white; width: 240px; border-radius: 12px; overflow: hidden; box-shadow: 0 3px 10px #
ddc8b8; } .price { color: #218044; font-weight: bold; font-size: 20px; }
```

5. CASE STUDY 3: ROYAL CAFE & RESTAURANT WEBPAGE (CASE20)

Case 20 - Royal Cafe & Restaurant

Full Interactive Web Application

Description: A comprehensive restaurant web portal incorporating a complete menu gallery, interactive table reservation form, customer review cards, and detailed opening schedules. Demonstrates complex form controls including text fields, selects, and textareas.

- **Reservation Form:** Styled inputs for date, time, party size, and special requests.
- **Customer Feedback:** Star rating cards with customized theme accents (`#fff4e5`).
- **Cohesive Theme:** Earthy brown and gold palette suited for hospitality branding.

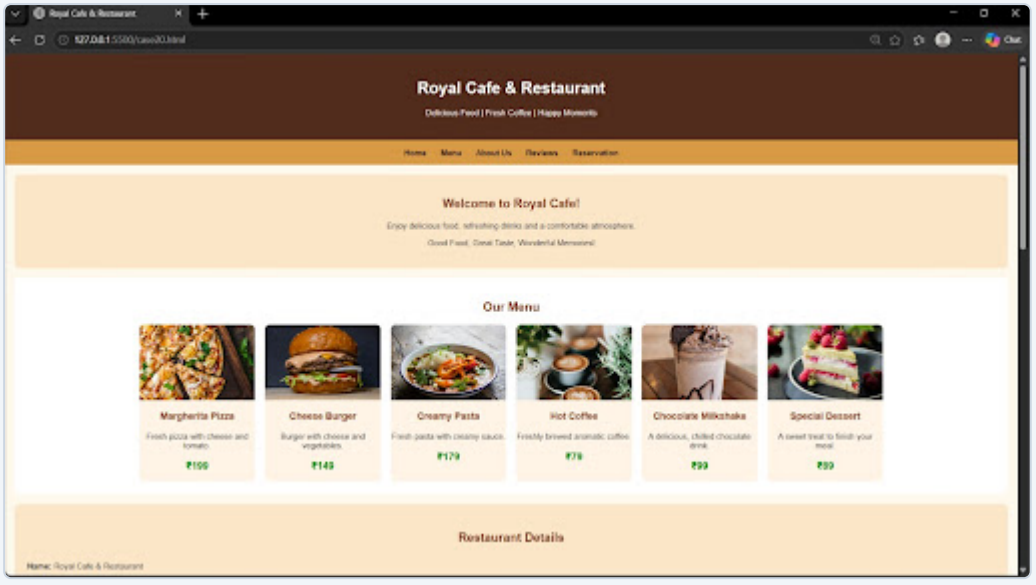


Figure 3.1: Web Preview Output for Case 20 (Royal Cafe & Restaurant Webpage)

6. COMPARATIVE PROJECT MATRIX

Case File	Target Domain	Key Layout Technique	Interactive Components
case1	Personal Portfolio	Block Sections & Inline Tags	Anchor Navigation Links
case9	Bakery E-Commerce	Flexbox Wrapped Grid	Order Buttons & Offers Banner
case20	Restaurant Portal	Multi-Column Grid & Forms	Table Reservation Form System